

NEA Implemented Text Revisions

All non-CAD, non-image, and non-physical test improvements consolidated into a single implementation guide.

Page 2 – Problem Identification

Fix: guitar sales have been on the rise and have increased by 6.4% in the past year alone according to Fender's 2024 sales report.

Note: Replace the 1998 NIH statistic with a post-2015 source if possible.

Page 4 – Design Brief

Replace: I will be designing a product that satisfies the needs and demands of middle and high school robotics enthusiasts. This product will be used over a period of approximately two weeks per module, making it essential that the design is self-contained and requires no specialist tools beyond those included in the kit.

Fix: The finish will be matte to give it a bolder appearance and reduce glare during assembly.

Page 6 – Client Profile

Add: Design implications: My client's workspace is limited in size. All parts must fit within a 200mm × 200mm × 200mm footprint when assembled.

Add: Mood board categories should explicitly reference F1 cars, military hardware, educational kits, and humanoid robots.

Page 7 – Aesthetics & Cost

Fix: The use of the ultrasound sensors as eyes makes the robot appear more human, forming a stronger connection between the user and the robot.

Fix: Use wording such as: blue is commonly associated with technology and engineering products.

Page 8 – Sustainability

Add: Applying the 6Rs to this product specifically: rethink, refuse, reduce, reuse, repair, recycle.

Fix: Describe PLA as industrially compostable under controlled conditions.

Page 9–11 – Product Analysis

Add: Include design implications from customer reviews, especially vibration-resistant fasteners and included tools.

Fix: Use metric units consistently throughout the NEA.

Page 12–14 – Manufacturing Research

Add: Explicitly state which manufacturing tools are actually used in the project and why.

Fix: A strip heater bends thermoplastics; it does not cut them.

Page 15 – Materials

Add: PLA selected due to low cost, ease of FDM printing, and ability to rapidly prototype replacement parts.

Fix: Oak, pine, MDF, and acrylic excluded due to weight, brittleness, or dimensional inconsistency.

Page 16 – Material Testing

Add: Conclusion: PLA remains the preferred chassis material due to digital tolerance control and reprintability.

Page 18 – Safety / Size / Function

Fix: Replace 'double insulation' with 'protected low-voltage wiring'.

Fix: Use 'designed with reference to CE and BSI safety guidance' instead of claiming certification.

Page 21 – Ergonomics

Fix: Differentiate between motor mechanical noise and electromagnetic coil whine.

Add: Include QR-code based digital instruction manual reference.

Page 22 – Specification

Fix: Budget / Cost should replace the incorrect Packaging heading.

Add: 1.1 should include a measurable 75% engagement survey target.

Add: 4.1 should specify at least 60% recyclable or biodegradable chassis material.

Page 23–25 – Initial Ideas

Add: Add 'Teaches:' annotations below every idea.

Fix: Expand weak cons with specific engineering limitations.

Page 26 – Scoring Matrix

Fix: Replace '1,2,9 and 10' with '1,2,9 and 12'.

Add: Explain that scores below 3 indicate failure to meet specification requirements.

Page 31 – Proposed Solution

Replace: I chose the bipedal robot (Idea 12) as my proposed design because it offers the strongest progression pathway from beginner to advanced robotics concepts.

Page 32–36 – Development

Add: Convert development paragraphs into engineering problem → cause → fix bullet lists.

Add: Document V1, V1.5, and V2 progression clearly.

Page 37 – Future Development

Add: Include future hardware upgrades such as STM32F4 support and Bluetooth expansion.

Add: Include full specification evaluation table.

Manual Changes Still Required

CAD: Add packaging column to scoring matrix.

CAD: Add legends to specification colour bars.

CAD: Add annotated arrows to CAD renders.

Images: Add captions to all prototype photographs.

Tests: Add PLA material testing and balancing photographs.

Tests: Conduct 10-user engagement survey for specification 1.1.